2024-25

S6 MATH EP

M1

CUHKFAA CHAN CHUN HA SECONDARY SCHOOL

2024-2025 Secondary Six First Term Examination

MATHEMATICS Extended Part

Module 1 (Calculus and Statistics)

Question-Answer Book

11/11/2024

8:15 am – 10:45 am (2½ hours)

This paper must be answered in English

Name: _____

Class: _____( )

	Marker Use Only
Question	Mark
1	
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Total			

INSTRUCTIONS

1. After the announcement of the start of the examination, you should first write your Name, Class and Class number in the spaces provided on Page 1.
2. Attempt **ALL** questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
3. Graph paper and supplementary answer sheets will be supplied on request. Write your Name, Class and Class number and mark the question number box on each sheet, and fasten them with string **INSIDE** this book.
4. Unless otherwise specified, all working must be clearly shown.
5. Unless otherwise specified, numerical answers should be either exact or given to 4 decimal places.
6. For definite integrals, answers obtained by using numerical integration functions in calculators are not accepted.
7. The diagrams in this paper are not necessarily drawn to scale.
8. No extra time will be given to candidates for writing your Name, Class and Class number or filling in the question number boxes after 'Time is up' announcement.

Section A (50 marks)

1. The table below shows the probability distribution of a discrete random variable X, where a and b are constants.

x	1	a	b	9
$P(X = x)$	0.2	0.3	0.1	0.4

It is given that $E(X) = 5.2$.

(a) Express b in terms of a.

(b) It is given that $\text{Var}(3X - 1) = 96.84$. Find a and b.

(5 marks)

Answers written in the margins will not be marked.

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2. Let A and B be two events. Denote the complementary event of B by B'. It is given that $P(A \cup B) = \frac{4}{3}P(A)$, $P(B) = \frac{1}{2}P(A)$ and $P(A \mid B') = k$, where k is a non-zero constant.

(a) Express $P(A)$ in terms of k.

(b) Are A and B mutually exclusive? Explain your answer.

(c) If A and B are independent, find the value of k. (6 marks)

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3. Define $f(x) = \dfrac{\ln x}{2\ln x - x}$ for all positive numbers x.

(a) Find the minimum value of $f(x)$.

(b) Find the coordinates of the point(s) of intersection of the curve $y = f(x)$ and the x-axis. Hence, find the equation of the tangent to the curve at each point of intersection.

(7 marks)

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4. Let k be a constant.

(a) Expand e^{-3kx} in ascending powers of x as far as the term in x^2.

(b) Let $y = 2e^{kx}$. If the coefficient of x^2 in the expansion of $\frac{(1+2x)^8}{y^3}$ is greater than the coefficient of x by 3, find k.

(6 marks)

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5. The probability that a day is a rainy day is $\frac{1}{9}$. Given that it is not a rainy day, the probability that John has an umbrella with him on that day is $\frac{1}{4}$. Given that John does not have an umbrella with him on a day, the probability that it is a rainy day is $\frac{1}{61}$.

(a) Find the probability that it is not a rainy day and John does not have an umbrella with him on that day.

(b) Given that it is a rainy day, find the probability that John has an umbrella with him on that day.

(c) Given that John has an umbrella with him on a day, find the probability that it is not a rainy day.

(7 marks)

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6. Let H cm be the height of a member in an organization. It is given that the standard deviation of H is 2.5. A manager wants to estimate μ, the mean of H. The manager randomly selects 100 members, and finds that the sum of the heights of these 100 members is 16 200 cm.

(a) Construct a 97% confidence interval for μ.

(b) Another random sample of 200 members is taken and a β% confidence interval for μ is constructed as follows:

(161.18 , 161.82).

Find β correct to the nearest integer.

(5 marks)

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7. Let $f(x)$ be a function such that $f'(x) = x(x + k^{x^2})$, where k is a positive constant. It is given that $-3f'(-1) = 2f'(1)$.

(a) Prove that $k = 5$.

(b) If the graph of $y = f(x)$ passes through the origin, find $f(x)$.

(6 marks)

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8. Let $f(x) = e^{-x^{\frac{1}{4}}}$.

(a) Using integration by substitution, find $\int x^{-\frac{3}{4}} f(x)dx$.

(b) Let $g(u) = e^{-u}(u^3 + 3u^2 + 6u + 3)$, where $u = x^{\frac{1}{4}}$. It is known that $\frac{dg(u)}{dx} = h(x) \times f(x)$.

Find the expression $h(x)$.

(c) Find the exact value of the area of the region bounded by the curve $y = f(x)$, the x-axis, the y-axis and the line $x = 16$.

(8 marks)

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Section B (50 marks)

9. Victoria is a salesperson of a company. Her monthly income is the sum of the fixed basic monthly salary plus a commission on sales generated each month. The basic monthly salary of Victoria is \$8 500 and the amount of her commission in a month follows a normal distribution with a mean of \$8 100 and a standard deviation of \$1 200. The income level of a salesperson in the company in a month is graded as low, normal and high according to the following table.

Monthly income (\$$x$)	Income level in a month
$x < 14\,800$	Low
$14\,800 \leq x \leq 17\,500$	Normal
$x > 17\,500$	High

(a) Find the probability that the income level of Victoria in a certain month is graded as high. (2 marks)

(b) Find the probability that the income level of Victoria is graded as high in fewer than 3 months and low in at most 1 month in the coming 6 months. (3 marks)

(c) Brian is also a salesperson in the company. The probabilities for his income level in a month to be graded as low and high are 0.15 and 0.25 respectively. It is known that the monthly incomes of Victoria and Brian are independent of each other.

(i) Find the probability that the income levels of Victoria and Brian are the same in a certain month.

(ii) Find the probability that the income levels of Victoria and Brian are the same for the 4th time in May starting from January of the same year.

(iii) The basic monthly salary of Victoria is adjusted. At least how much should be her new basic monthly salary such that the probability for her income level in a month to be graded as high is not less than Brian? (Give your answer correct to the nearest hundred dollars.)

(7 marks)

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10. The number of parking penalty notices issued to the owners of the cars on a certain road on a day follows a Poisson distribution with a mean of 2.2. If the number of parking penalty notices issued on a day is less than 3, that day is regarded as *good*.

(a) Find the probability that a certain day is *good*. (3 marks)

(b) Find the probability that more than 2 days are *good* in 5 consecutive days. (3 marks)

(c) Find the probability that, in 3 consecutive days, all the 3 days are *good* and the total number of parking penalty notices issued in these 3 days is 4. (3 marks)

(d) Given that the total number of parking penalty notices issued in the first 3 days of 5 consecutive days is 4, find the probability that exactly 3 of the 5 consecutive days are *good*. (4 marks)

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11. A researcher models the rate of change of the number of births (in thousand per year) of a city by

$$B(t) = 9e^{-0.01t^2+0.2t-1},$$

where t (≥ 0) is the number of years elapsed since the research starts. Suppose there are α thousand births in the first 2 years since the research starts. Let α_1 be the estimate of α by using the trapezoidal rule with 5 sub-intervals.

(a) (i) Find α_1.

(ii) Determine whether α_1 is an over-estimate or an under-estimate. Explain your answer. (6 marks)

(b) In the same research, the researcher models the rate of change of the number of deaths (in thousand per year) of the city by

$$D(t) = \frac{15(2t+1)\ln(2t^2+2t+4)}{4t^2+4t+8},$$

where t (≥ 0) is the number of years elapsed since the research starts. Suppose there are β thousand deaths in the first 2 years since the research starts.

(i) Find β.

(ii) The researcher claims that $\frac{\beta-\alpha}{\alpha} > \frac{1}{3}$. Do you agree? Explain your answer.

(6 marks)

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12. A biologist, Sara, studies the growth of algae on the surface of a pond such that the area A (in thousand m^2) of the pond covered by the algae is given

$$A = \frac{50a^{kt-2}}{3a^{kt-2}+1},$$

where t ($t \geq 0$) is the number of months elapsed since the start of the study, and a and k are constants.

(a) Express $\ln\left(\frac{50}{A}-3\right)$ as a linear function of t. (2 marks)

(b) Sara finds that the slope and the intercept on the horizontal axis of the linear function in (a) are $-\ln\sqrt[5]{3}$ and 10 respectively.

(i) Find the values of a and k.

(ii) Find $\frac{dA}{dt}$ and $\frac{d^2A}{dt^2}$.

(iii) Will the area of the pond covered by the algae exceed 17 000 m^2? Explain your answer.

(iv) Find the area of the pond covered by the algae when its rate of change is a maximum.

(11 marks)

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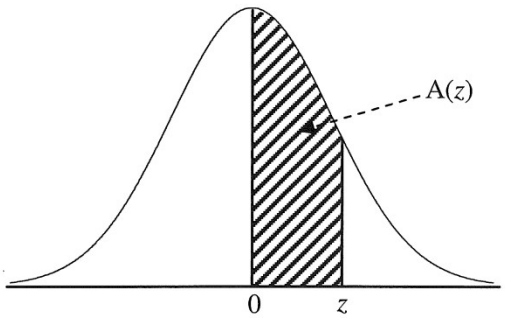
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End of Paper

Table: Area under the Standard Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995
3.3	.4995	.4995	.4995	.4996	.4996	.4996	.4996	.4996	.4996	.4997
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4998
3.5	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998

Note : An entry in the table is the proportion of the area under the entire curve which is between $z = 0$ and a positive value of z. Areas for negative values of z are obtained by symmetry.



$$A(z) = \int_0^z \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} dx$$